

\*  
\* IC Technology: CMOS 0.18µm - 6 Metal  
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VDD 1 0 DC 2.00

Vclk 117 0 DC 0 PULSE(0.00 2.00 0.50N 0.05N 0.05N 0.50N 1.10N)

\*  
\* List of nodes

\* "\_qb7" corresponds to n°3  
\* "\_q7" corresponds to n°5  
\* "\_n4\_7" corresponds to n°6  
\* "\_n3\_7" corresponds to n°7  
\* "\_n2\_7" corresponds to n°8  
\* "\_n1\_7" corresponds to n°9  
\* "\_qb6" corresponds to n°10  
\* "\_q6" corresponds to n°11  
\* "\_n4\_6" corresponds to n°12  
\* "\_n3\_6" corresponds to n°13  
\* "\_n2\_6" corresponds to n°14  
\* "\_n1\_6" corresponds to n°15  
\* "\_qb5" corresponds to n°16  
\* "\_q5" corresponds to n°17  
\* "\_n4\_5" corresponds to n°18  
\* "\_n3\_5" corresponds to n°19  
\* "\_n2\_5" corresponds to n°20  
\* "\_n1\_5" corresponds to n°21  
\* "\_qb4" corresponds to n°22  
\* "\_q4" corresponds to n°23  
\* "\_n4\_4" corresponds to n°24  
\* "\_n3\_4" corresponds to n°25  
\* "\_n2\_4" corresponds to n°26  
\* "\_n1\_4" corresponds to n°27  
\* "\_qb3" corresponds to n°28  
\* "\_q3" corresponds to n°29  
\* "\_n4\_3" corresponds to n°30  
\* "\_n3\_3" corresponds to n°31  
\* "\_n2\_3" corresponds to n°32  
\* "\_n1\_3" corresponds to n°33  
\* "\_qb2" corresponds to n°34  
\* "\_q2" corresponds to n°35  
\* "\_n4\_2" corresponds to n°36  
\* "\_n3\_2" corresponds to n°37  
\* "\_n2\_2" corresponds to n°38  
\* "\_n1\_2" corresponds to n°39  
\* "\_qb1" corresponds to n°40  
\* "\_q1" corresponds to n°41  
\* "\_n4\_1" corresponds to n°42  
\* "\_n3\_1" corresponds to n°43  
\* "\_n2\_1" corresponds to n°44  
\* "\_n1\_1" corresponds to n°45  
\* "\_qb0" corresponds to n°46  
\* "\_q0" corresponds to n°47  
\* "\_n4\_0" corresponds to n°48  
\* "\_n3\_0" corresponds to n°49  
\* "\_n2\_0" corresponds to n°50  
\* "\_n1\_0" corresponds to n°51  
\* "\_feedback" corresponds to n°52  
\* "\_x2" corresponds to n°55  
\* "\_x1" corresponds to n°58  
\* "N100" corresponds to n°100  
\* "N101" corresponds to n°101  
\* "N102" corresponds to n°102  
\* "N103" corresponds to n°103  
\* "N104" corresponds to n°104  
\* "N105" corresponds to n°105  
\* "N106" corresponds to n°106  
\* "N107" corresponds to n°107  
\* "N108" corresponds to n°108  
\* "N109" corresponds to n°109  
\* "N110" corresponds to n°110  
\* "N111" corresponds to n°111  
\* "N112" corresponds to n°112  
\* "N113" corresponds to n°113  
\* "N114" corresponds to n°114  
\* "N115" corresponds to n°115  
\* "N116" corresponds to n°116  
\* "clk" corresponds to n°117  
\*

\* MOS devices

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MN2 0 5 61 0 N1 W= 0.60U L= 0.20U

MN3 62 3 5 0 N1 W= 0.60U L= 0.20U

MN4 0 8 62 0 N1 W= 0.60U L= 0.20U

MN5 63 11 6 0 N1 W= 0.60U L= 0.20U

MN6 0 7 63 0 N1 W= 0.60U L= 0.20U  
MN7 64 6 7 0 N1 W= 0.60U L= 0.20U  
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MN9 0 8 65 0 N1 W= 0.60U L= 0.20U  
MN10 66 117 8 0 N1 W= 0.60U L= 0.20U  
MN11 0 9 66 0 N1 W= 0.60U L= 0.20U  
MN12 67 6 9 0 N1 W= 0.60U L= 0.20U  
MN13 0 8 67 0 N1 W= 0.60U L= 0.20U  
MN14 68 13 10 0 N1 W= 0.60U L= 0.20U  
MN15 0 11 68 0 N1 W= 0.60U L= 0.20U  
MN16 69 10 11 0 N1 W= 0.60U L= 0.20U  
MN17 0 14 69 0 N1 W= 0.60U L= 0.20U  
MN18 70 17 12 0 N1 W= 0.60U L= 0.20U  
MN19 0 13 70 0 N1 W= 0.60U L= 0.20U  
MN20 71 12 13 0 N1 W= 0.60U L= 0.20U  
MN21 72 117 71 0 N1 W= 0.60U L= 0.20U  
MN22 0 14 72 0 N1 W= 0.60U L= 0.20U  
MN23 73 117 14 0 N1 W= 0.60U L= 0.20U  
MN24 0 15 73 0 N1 W= 0.60U L= 0.20U  
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MN26 0 14 74 0 N1 W= 0.60U L= 0.20U  
MN27 75 19 16 0 N1 W= 0.60U L= 0.20U  
MN28 0 17 75 0 N1 W= 0.60U L= 0.20U  
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MN30 0 20 76 0 N1 W= 0.60U L= 0.20U  
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MN38 81 18 21 0 N1 W= 0.60U L= 0.20U  
MN39 0 20 81 0 N1 W= 0.60U L= 0.20U  
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MN41 0 23 82 0 N1 W= 0.60U L= 0.20U  
MN42 83 22 23 0 N1 W= 0.60U L= 0.20U  
MN43 0 26 83 0 N1 W= 0.60U L= 0.20U  
MN44 84 29 24 0 N1 W= 0.60U L= 0.20U  
MN45 0 25 84 0 N1 W= 0.60U L= 0.20U  
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MN47 86 117 85 0 N1 W= 0.60U L= 0.20U  
MN48 0 26 86 0 N1 W= 0.60U L= 0.20U  
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MN50 0 27 87 0 N1 W= 0.60U L= 0.20U  
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MN65 0 32 95 0 N1 W= 0.60U L= 0.20U  
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MN69 0 38 97 0 N1 W= 0.60U L= 0.20U  
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MN71 0 37 98 0 N1 W= 0.60U L= 0.20U  
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MN80 0 41 103 0 N1 W= 0.60U L= 0.20U  
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MN82 0 44 104 0 N1 W= 0.60U L= 0.20U  
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MN85 106 42 43 0 N1 W= 0.60U L= 0.20U  
MN86 107 117 106 0 N1 W= 0.60U L= 0.20U  
MN87 0 44 107 0 N1 W= 0.60U L= 0.20U  
MN88 108 117 44 0 N1 W= 0.60U L= 0.20U  
MN89 0 45 108 0 N1 W= 0.60U L= 0.20U

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MN91 0 44 109 0 N1 W= 0.60U L= 0.20U  
MN92 110 49 46 0 N1 W= 0.60U L= 0.20U  
MN93 0 1 110 0 N1 W= 0.60U L= 0.20U  
MN94 111 46 1 0 N1 W= 0.60U L= 0.20U  
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MN96 112 52 48 0 N1 W= 0.60U L= 0.20U  
MN97 0 49 112 0 N1 W= 0.60U L= 0.20U  
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MP2 1 5 3 1 P1 W= 1.60U L= 0.20U  
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MP100 1 50 49 1 P1 W= 1.60U L= 0.20U  
MP101 50 117 1 1 P1 W= 1.60U L= 0.20U  
MP102 1 51 50 1 P1 W= 1.60U L= 0.20U  
MP103 51 48 1 1 P1 W= 1.60U L= 0.20U  
MP104 1 50 51 1 P1 W= 1.60U L= 0.20U  
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MP108 1 56 55 1 P1 W= 1.60U L= 0.20U  
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MP110 1 58 57 1 P1 W= 1.60U L= 0.20U  
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MP112 60 23 59 1 P1 W= 1.60U L= 0.20U  
MP113 1 29 60 1 P1 W= 1.60U L= 0.20U  
\*

C2 1 0 391.019fF  
C3 3 0 4.410fF  
C5 5 0 17.323fF  
C6 6 0 6.670fF  
C7 7 0 6.772fF  
C8 8 0 7.591fF  
C9 9 0 4.022fF  
C10 10 0 4.410fF  
C11 11 0 18.003fF  
C12 12 0 6.670fF  
C13 13 0 6.772fF  
C14 14 0 7.591fF  
C15 15 0 4.022fF  
C16 16 0 4.410fF  
C17 17 0 16.725fF  
C18 18 0 6.670fF  
C19 19 0 6.772fF  
C20 20 0 7.591fF  
C21 21 0 4.022fF  
C22 22 0 4.410fF  
C23 23 0 15.161fF  
C24 24 0 6.670fF  
C25 25 0 6.772fF  
C26 26 0 7.591fF  
C27 27 0 4.022fF  
C28 28 0 4.410fF  
C29 29 0 14.483fF  
C30 30 0 6.670fF  
C31 31 0 6.772fF  
C32 32 0 7.591fF

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C34 34 0 4.410fF  
C35 35 0 12.631fF  
C36 36 0 6.670fF  
C37 37 0 6.772fF  
C38 38 0 7.591fF  
C39 39 0 4.022fF  
C40 40 0 4.410fF  
C41 41 0 11.135fF  
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C46 46 0 4.410fF  
C47 1 0 9.639fF  
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C72 72 0 0.308fF  
C73 73 0 0.308fF  
C74 74 0 0.308fF  
C75 75 0 0.308fF  
C76 76 0 0.308fF  
C77 77 0 0.308fF  
C78 78 0 0.308fF  
C79 79 0 0.308fF  
C80 80 0 0.308fF  
C81 81 0 0.308fF  
C82 82 0 0.308fF  
C83 83 0 0.308fF  
C84 84 0 0.308fF  
C85 85 0 0.308fF  
C86 86 0 0.308fF  
C87 87 0 0.308fF  
C88 88 0 0.308fF  
C89 89 0 0.308fF  
C90 90 0 0.308fF  
C91 91 0 0.308fF  
C92 92 0 0.308fF  
C93 93 0 0.308fF  
C94 94 0 0.308fF  
C95 95 0 0.308fF  
C96 96 0 0.308fF  
C97 97 0 0.308fF  
C98 98 0 0.308fF  
C99 99 0 0.308fF  
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C104 104 0 0.308fF  
C105 105 0 0.308fF  
C106 106 0 0.308fF  
C107 107 0 0.308fF  
C108 108 0 0.308fF  
C109 109 0 0.308fF  
C110 110 0 0.308fF  
C111 111 0 0.308fF  
C112 112 0 0.308fF  
C113 113 0 0.308fF  
C114 114 0 0.308fF  
C115 115 0 0.308fF  
C116 116 0 0.308fF

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C117 117 0 32.839fF
*
*
* n-MOS BSIM4 :
* low leakage
.MODEL N1 NMOS LEVEL=14 VTHO=0.55 U0=0.038 TOXE= 4.0E-9 LINT=-0.010U
+K1 =0.170 K2=0.100 DVT0=2.300
+DVT1=0.540 LPE0=23.000e-9 ETA0=0.080
+NFACTOR= 1.7 U0=0.038 UA=2.800e-15
+WINT=0.020U LPE0=23.000e-9
+KT1=-0.060 UTE=-1.800 VOFF=0.010
+XJ=0.150U NDEP=170.000e15 PCLM=0.290
+CGSO=100.0p CGDO=100.0p
+CGBO= 60.0p
*
* p-MOS BSIM4:
* low leakage
.MODEL P1 PMOS LEVEL=14 VTHO=-0.60 U0=0.010 TOXE= 4.0E-9 LINT=-0.040U
+K1 =0.290 K2=0.100 DVT0=2.300
+DVT1=0.540 LPE0=23.000e-9 ETA0=0.080
+NFACTOR= 2.2 U0=0.010 UA=1.000e-15
+WINT=0.020U LPE0=23.000e-9
+KT1=-0.060 UTE=-1.800 VOFF=0.010
+XJ=0.150U NDEP=170.000e15 PCLM=0.300
+CGSO=100.0p CGDO=100.0p
+CGBO= 60.0p
*
* Transient analysis
*
* (Winspice)
.options temp=27.0
.control
tran 0.1N 5.00N
print V(29) V(23) V(17) V(5) V(117) V(47) V(41) V(35) V(11) > out.txt
plot V(29) V(23) V(17) V(5) V(117) V(47) V(41) V(35) V(11)
.endc
.END

```